Stibnite

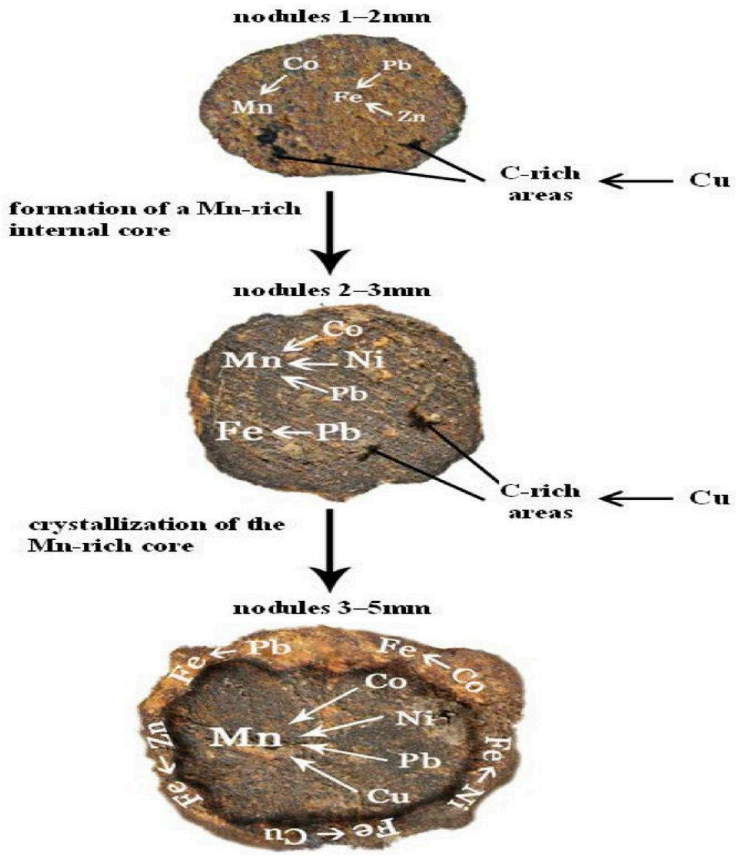
Stibnite crystals

Characteristics - Stibnite, also known as antimonite, is a sulfide mineral. This crystalline solid is a mineral form of antimony trisulfide (Sb_2S_3) and is characterized by its soft, metallic grey appearance and orthorhombic space group. It serves as the primary source for the metalloid antimony.

Natural occurrence - Stibnite may occur in a variety of hydrothermal mineral deposits, but is most frequently found in low-temperature assemblages where it may be associated with galena, pyrite, antimony sulphosalts and gold. It may also occur in hot springs,epithermal deposits

Fun Facts - The ancient Egyptians used powdered stibnite as kohl eyeliner .

Manganese Nodules



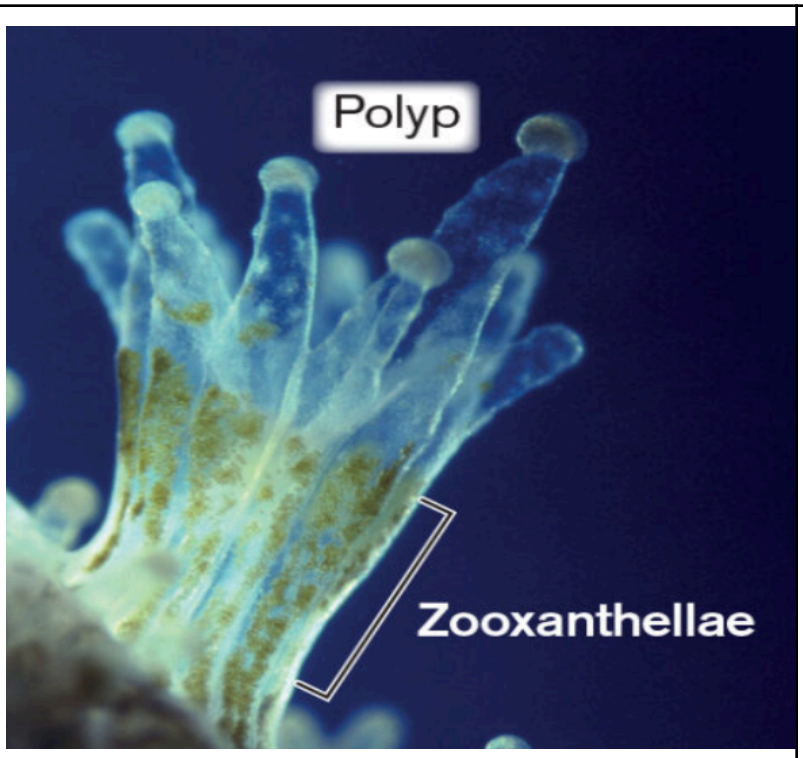
Process of formation of manganese nodules

Characteristic - Manganese nodules are mineral structures on the sea bottom formed of concentric layers of iron and manganese hydroxides around a core. Such nodules have sizes ranging from 3 to 10 cm. They can have various shapes from spherical ,elliptical or irregular with most of them also having botryoidal texture (like a bunch of grapes) .

Natural occurrence - Nodules often lie on the sea bed partly or completely embedded in soft pelagic sediments. They cover 70% of the seabed, especially abyssal plains of deep oceans . They are also found in shallower waters and are even found in lakes

Fun facts - Did u know that manganese nodules usually formed around a nucleus and that nucleus can be anything from a shell piece to volcanic fragment and even sometimes shark tooth .

Coral



Microscopic view of a singular polyp

Characteristics- Corals are made of a colony of polyps, tubular sac like animals with a central mouth surrounded by a ring of tentacles. Such polyps can vary in size from several inches to centimeters . These colonies can rise to impressive heights of 13 feet corals can be of various colors of red, blue, green . They are filter feeders using small microscopic tentacles to catch food floating around themselves .Zooxanthellae are photosynthetic algae living inside coral tissues in a symbiotic relationship. They provide up to 90% of the coral's energy by producing sugars and amino acids through photosynthesis, help corals build calcium carbonate skeletons for reef formation,

Natural Occurrence - Corals form large systems called reefs which are mostly formed in shallow shelf environments in tropical-subtropical oceans. In rare cases some coral species do live in deep cooler oceans.

Fun facts - Coral polyps form structures called reefs, which are the largest living structures on Earth. The most famous of these is the Great Barrier Reef, stretching about 2,300 km in length and even visible from space. .

Brachiopods

Extant brachiopods in their natural habitat

Characteristics- Brachiopods are marine invertebrates having a shell similar to that of bivalves. Unlike bivalves they have a line of symmetry splitting the valves down the middle instead of along the commissure. Brachiopods mostly bind to the sea bed using a fleshy stalk called a pedicle. They filter food using a ciliated tentacle called lophophore.

Natural occurrence – They are mostly found in a variety of marine environments from tropical to polar oceans. They are rarely found at turbulent coastlines where bivalves such as clams dominate. Instead they are found in deep, colder and steady water where they live attached to rocks, shells, or the seafloor

Fun fact –While externally they closely resemble bivalves. Taxonomically they are not classified under bivalvia. In fact they are not even classified under mollusca and are part of their own separate group**.**